

Kaiqi Yang

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Research Interests

- Applications of Large Language Models; Conversational AI; Computational Social Science

Education

- **Michigan State University** 2022 – 2027 (*expected*)
PhD Student, Computer Science and Engineering East Lansing, USA
 - Advisor: Dr. Hui Liu, Dr. Jiliang Tang
- **Fudan University** 2019 – 2021
Master of Applied Statistics, School of Data Science Shanghai, China
- **Fudan University** 2015 – 2019
Bachelor of Sociology, Department of Sociology Shanghai, China

Experiences

- **Data Science and Engineering Lab, Michigan State University** Aug 2022 – Present
Research Assistant, Department of Computer Science and Engineering East Lansing, USA
 - Research Areas: Large Language Models, AI for Education, Computational Social Science
 - Advisors: Dr. Jiliang Tang and Dr. Hui Liu
- **Amazon Inc.** Aug 2025 – Nov 2025
Applied Scientist Intern Seattle, USA
 - Research Topics: Foundational AI for Stores; Conversational Generation and Evaluation
- **TAL Education Group** Jun 2024 – Aug 2024
Summer Intern Beijing, China
 - Research Topics: Applications of Large Language Models; Intelligent Tutoring Systems for Elementary Math
- **Department of Computing, The Hong Kong Polytechnic University** May 2023 – Aug 2023
Research Intern Hong Kong
 - Research Topics: Large Language Models; Graph Data Mining

Publications

* indicates equal contribution.

- *LLM-based Automated Grading with Human-in-the-Loop*
Hang Li*, Yucheng Chu*, **Kaiqi Yang**, Yasemin Copur-Gencturk, Jiliang Tang
IEEE International Conference on Teaching, Assessment, and Learning for Engineering (TALE), 2025
- *Iterative LLM-Based Generation and Refinement of Distracting Conditions in Math Word Problems*
Kaiqi Yang, Hang Li, Yucheng Chu, Zitao Liu, Mi Tian, Hui Liu
IEEE International Conference on Teaching, Assessment, and Learning for Engineering (TALE), 2025
- *Ask-Before-Detection: Identifying and Mitigating Conformity Bias in LLM-Powered Error Detector for Math Word Problem Solutions*
Hang Li, Tianlong Xu, **Kaiqi Yang**, Yucheng Chu, Yanling Chen, Yichi Song, Qingsong Wen, Hui Liu
Annual Meeting of the Association for Computational Linguistics (ACL), 2025
- *Are Large Language Models (LLMs) Good Social Predictors?*
Kaiqi Yang, Hang Li, Hongzhi Wen, Tai-Quan Peng, Jiliang Tang, Hui Liu
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2024
- *Content Knowledge Identification with Multi-Agent Large Language Models (LLMs)*
Kaiqi Yang, Yucheng Chu, Taylor Darwin, Ahreum Han, Hang Li, Hongzhi Wen, Yasemin Copur-Gencturk, Jiliang Tang, Hui Liu
International Conference on Artificial Intelligence in Education (AIED), 2024
- *Spectral-Aware Augmentation for Enhanced Graph Representation Learning*
Kaiqi Yang, Haoyu Han, Wei Jin, Hui Liu
ACM International Conference on Information and Knowledge Management (CIKM), 2024

Preprints

* indicates equal contribution.

- *Optimizing In-Context Demonstrations for LLM-based Automated Grading*
Yucheng Chu, Hang Li, **Kaiqi Yang**, Yasemin Copur-Gencturk, Kevin Haudek, Joseph Krajcik, Jiliang Tang
arXiv preprint arXiv:2603.00465, 2026.
- *Confusion-Aware Rubric Optimization for LLM-based Automated Grading*
Yucheng Chu, Hang Li, **Kaiqi Yang**, Yasemin Copur-Gencturk, Joseph Krajcik, Namsoo Shin, Jiliang Tang
arXiv preprint arXiv:2603.00451, 2026.
- *How Uncertain Is the Grade? A Benchmark of Uncertainty Metrics for LLM-Based Automatic Assessment*
Kaiqi Yang*, Hang Li*, Xianxuan Long, Fedor Filippov, Yucheng Chu, Yasemin Copur-Gencturk, Peng He, Cory Miller, Namsoo Shin, Joseph Krajcik, Hui Liu, Jiliang Tang
arXiv preprint arXiv:2602.16039, 2026.
- *Exploring Solution Divergence and Its Effect on Large Language Model Problem Solving*
Hang Li, **Kaiqi Yang**, Yucheng Chu, Hui Liu, Jiliang Tang
arXiv preprint arXiv:2509.22480, 2025.
- *A LLM-Driven Multi-Agent Systems for Professional Development of Mathematics Teachers*
Kaiqi Yang*, Hang Li*, Yucheng Chu, Ahreum Han, Yasemin Copur-Gencturk, Jiliang Tang, Hui Liu
arXiv preprint arXiv:2507.05292, 2025.

- *Enhancing LLM-Based Short Answer Grading with Retrieval-Augmented Generation*
Yucheng Chu, Peng He, Hang Li, Haoyu Han, **Kaiqi Yang**, Yu Xue, Tingting Li, Joseph Krajcik, Jiliang Tang
arXiv preprint arXiv:2504.05276, 2025.
- *Beyond Partisan Leaning: A Comparative Analysis of Political Bias in Large Language Models*
Kaiqi Yang, Hang Li, Yucheng Chu, Yuping Lin, Tai-Quan Peng, Hui Liu
arXiv preprint arXiv:2412.16746, 2024.
- *Exploring Social Desirability Response Bias in Large Language Models: Evidence from GPT-4 Simulations*
Sanguk Lee*, **Kaiqi Yang***, Tai-Quan Peng, Ruth Heo, Hui Liu
arXiv preprint arXiv:2410.15442, 2024.
- *A LLM-Powered Automatic Grading Framework with Human-Level Guidelines Optimization*
Yucheng Chu, Hang Li, **Kaiqi Yang**, Harry Shomer, Hui Liu, Yasemin Copur-Gencturk, Jiliang Tang
arXiv preprint arXiv:2410.02165, 2024.

Presentations

- *Bringing Generative Artificial Intelligence (GenAI) to Education*
Hang Li, **Kaiqi Yang**, Yucheng Chu, Jiliang Tang
Tutorial in The 18th International Conference on Educational Data Mining (EDM), Palermo, Italy, 2025.
- *Innovative Artificial Intelligence Solutions for Improving Online Professional Development for Teachers: Multi-Agent Large Language Models*
Kaiqi Yang, Yasemin Copur-Gencturk, Ahreum Han, Hang Li, Jingxian Li, Jiliang Tang
Symposium “The Use of AI in Education and Research in Meaningful and Responsible Ways”
Annual Meeting of the American Educational Research Association (AERA), Denver, USA, 2025.

Honors and Awards

- Student Travel Award, ACM CIKM 2024
- Outstanding Graduate Student Scholarship, School of Data Science, Fudan University 2020/21
- Outstanding Undergraduate Student Scholarship, Fudan University 2016/17/18/19

Teaching

- Teaching Assistant, CSE 482: Big Data Analysis Spring 2025, Spring 2024
Responsibilities included grading coursework, holding office hours, and mentoring students on honors projects.
- Teaching Assistant, CSE 440: Introduction to Artificial Intelligence Fall 2024, Fall 2023
Responsibilities included managing online discussions, holding office hours, and grading coursework.

Mentoring

- Yucheng Chu, PhD Student
Project: Large Language Models for STEM Education
Co-authored publication: AIED 2024

Dec 2023 – Jan 2025

Service

- Conference Reviewer
Reviewed for WWW, WSDM, NeurIPS, ICLR, ICDM, ICML, ACL
- Journal Reviewer
Reviewed for TKDE, TNNLS, TKDD

2022 – Present

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